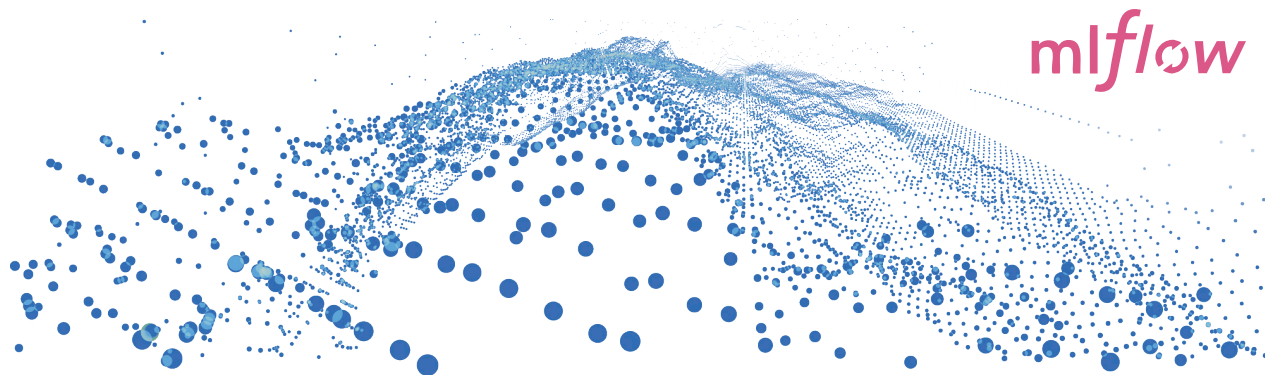


MLflow: Meeting the Challenges Posed by ML Based Application Development

Machine learning has revolutionised the manner in which software is built. However, there are many specific challenges associated with ML based application development. MLflow is an open source platform that offers solutions for these challenges. The major advantages of MLflow are the ability to work with any ML library, run the ML app the same way in any cloud, and scalability. This article introduces various features of MLflow.



Software development has undergone a revolutionary change due to machine learning. ML has made it possible to solve complex problems that were unsolvable with traditional approaches. However, the application development process faces newer challenges now, and the major ones are illustrated in Figure 1.

- **Difficulty in keeping track of experiments:** One of the major challenges of ML projects is keeping track of experiments. In traditional application development, the output is determined majorly by the code and the input. But the case with machine learning is different. Here, the output depends on code, data and the hyper-

parameters that are tweaked to get the optimal results. Hence, to track an experiment, the developer needs to track the combination of code, data and parameters involving the current experiment.

- **Difficulty in reproducing the working code:** In traditional application development the process revolves around one major language with minimal dependencies on external libraries.



Figure 1: Challenges due to machine learning

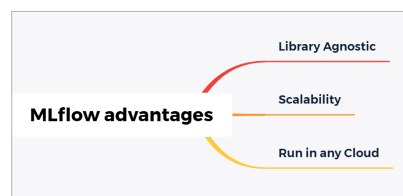


Figure 2: MLflow advantages